

NEXT.JS LOYIHASINI ahost.uz (CloudLinux Node.js Selector) GA DEPLOY QILISH – TO'LIQ KETMA-KETLIK

Ushbu qo'llanma har qanday Next.js (App Router yoki Pages Router) loyihasini cPanel + CloudLinux Node.js Selector orqali ishga tushirish uchun mo'ljallangan. ahost.uz, BlueHost, Namecheap, Hostinger (Business+) va boshqa CloudLinux'li shared hosting'larda ishlaydi.

TALABLAR

- Hosting tarifi Node.js 18+ ni qo'llab-quvvatlaydi (Next 14/15/16 uchun 20.x yoki 22.x tavsiya qilinadi).
- cPanel'da "Setup Node.js App" yoki "Node.js Selector" mavjud.
- Disk kvotasi kamida 500 MB bo'sh joy (node_modules uchun).
- RAM limiti kamida 512 MB (SSR uchun). Shared'da kam bo'lsa, VPS kerak bo'ladi.
- Backend alohida joylangan bo'lsa, uning URL'i tayyor.

1-QADAM: LOYIHANI STANDALONE OUTPUT'GA SOZLASH

next.config.js (yoki next.config.mjs) faylini oching va shu satrni qo'shing:

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  output: 'standalone',
  // ... boshqa sozlamalar
};
module.exports = nextConfig;
```

Nima uchun? "standalone" rejimi Next.js'ning kichik mustaqil serverini yaratadi – barcha kerakli fayllar bitta papkada, deploy paketi 10-20 marta kichik bo'ladi.

2-QADAM: ENVIRONMENT VARIABLES'NI TAYYORLASH

Loyiha ildizida .env (yoki .env.production) faylini yarating:

```
NEXT_PUBLIC_API_URL=https://api.example.com/api
# boshqa public env'lar (NEXT_PUBLIC_ prefiksi bilan)
```

MUHIM:

- "NEXT_PUBLIC_" prefiksli env'lar build paytida JS'ga jo'natiladi.
- Maxfiy kalitlarni (DATABASE_URL, JWT_SECRET, va h.k.) .env ga yozmang – ularni cPanel paneli orqali qo'shing (5-qadamga qarang).

3-QADAM: PRODUCTION BUILD QILISH

Terminal'da loyiha papkasida:

```
rm -rf .next deploy *.zip
npm install
npm run build
```

Build muvaffaqiyatli o'tsa, .next/standalone/ papkasi paydo bo'ladi.

4-QADAM: DEPLOY BUNDLE YIG'ISH

Terminal'da (Linux/macOS):

```
mkdir deploy
cp -R .next/standalone/. deploy/
cp -R .next/static deploy/.next/static
cp -R public deploy/public
cp .env deploy/.env
cp package-lock.json deploy/package-lock.json
rm -rf deploy/node_modules
zip -rq deploy.zip deploy
```

DIQQAT (CloudLinux talabi):

deploy/ papkasi ichida node_modules BO'LMASLIGI kerak!
Cloudlinux uni o'zi virtual environment'da yaratadi va symlink qiladi. Agar node_modules qoldirsangiz, "application root must not contain node_modules" xatosi chiqadi.

Windows'da (PowerShell):

```
Remove-Item -Recurse -Force .next, deploy, *.zip -ErrorAction Ignore
npm install
npm run build
New-Item -ItemType Directory deploy
Copy-Item -Recurse .next\standalone\* deploy\
Copy-Item -Recurse .next\static deploy\.next\static
Copy-Item -Recurse public deploy\public
Copy-Item .env deploy\.env
Copy-Item package-lock.json deploy\package-lock.json
Remove-Item -Recurse -Force deploy\node_modules -ErrorAction Ignore
Compress-Archive -Path deploy\* -DestinationPath deploy.zip
```

5-QADAM: cPanel'DA NODE.JS APP YARATISH

1. cPanel'ga kiring.
2. "Setup Node.js App" (yoki "Node.js Selector") ni oching.
3. "Create Application" tugmasini bosing.
4. Maydonlarni to'ldiring:

- Node.js version: 20.x yoki 22.x
- Application mode: Production
- Application root: myapp (loyiha papkasining nomi)
- Application URL: example.com (yoki subdomain)
- Application startup file: server.js
- Passenger log file: (bo'sh qoldiring yoki default)

5. "Create" ni bosing.

Endi cPanel /home/USERNAME/myapp/ papkasini yaratadi va CloudLinux virtual environment'ni sozlaydi.

6-QADAM: FAYLLARNI YUKLASH

Variant A – File Manager orqali:

1. cPanel → File Manager → /home/USERNAME/myapp/ ga kiring.
2. deploy.zip ni Upload qiling.
3. Faylni tanlab "Extract" tugmasini bosing.
4. Ichidagi "deploy/" papkasining barcha fayllarini (yashirin fayllar bilan birga) myapp/ ildiziga ko'chiring.
5. Bo'sh deploy/ papkasini va zip faylni o'chiring.

Variant B – SSH orqali (tezroq):

```
scp deploy.zip USERNAME@example.com:~/myapp.zip
ssh USERNAME@example.com
cd ~/myapp
unzip -o ~/myapp.zip
mv deploy/.[!~]* deploy/* . 2>/dev/null
rmdir deploy
rm ~/myapp.zip
```

Tekshirish: ildizda quyidagi fayllar bo'lishi kerak:

```
.env
.next/
package.json
package-lock.json
public/
server.js
```

node_modules KO'RINMASLIGI kerak (CloudLinux uni alohida virtual env'da yaratadi).

7-QADAM: ENVIRONMENT VARIABLES'NI cPanel'GA QO'SHISH

cPanel → Setup Node.js App → ilovangizni tanlang → "Environment variables" bo'limiga eng kamida shularni qo'shing:

```
NODE_ENV=production
HOSTNAME=0.0.0.0
NEXT_PUBLIC_API_URL=https://api.example.com/api
```

Maxfiy kalitlar bo'lsa shu yerga qo'shing:

```
DATABASE_URL=postgres://...
JWT_SECRET=...
STRIPE_SECRET_KEY=...
```

DIQQAT: PORT ni qo'lda kiritmang – CloudLinux o'zi beradi.

8-QADAM: NPM INSTALL

cPanel'da "Run NPM Install" tugmasini bosing.

Bu virtual env ichida `npm install --production` ishga tushiradi va package-lock.json'ga muvofiq dependency'larni o'rnatadi.

- Birinchi marta: 2-5 daqiqa.
- node_modules virtual env'da ~150-300 MB bo'ladi (loyihaga qarab).

Agar tugma ishlamasa, SSH orqali qo'lda:

```
source /home/USERNAME/nodeenv/myapp/20/bin/activate
cd ~/myapp
npm install --production
```

9-QADAM: ILOVANI ISHGA TUSHIRISH

cPanel → Setup Node.js App → "Restart" (yoki "Start App") tugmasi.

Loglarni ko'rish: shu sahifada "Show Logs" tugmasi yoki:

```
tail -f ~/myapp/stderr.log
tail -f ~/myapp/stdout.log
```

Agar xato bo'lsa, log'da aniq qator ko'rsatiladi.

10-QADAM: DOMEN VA SSL

1. Application URL maydonida ko'rsatgan domen avtomatik proxy qilinadi (Passenger orqali).
2. cPanel → SSL/TLS Status → "Run AutoSSL" – Let's Encrypt sertifikat bepul o'rnatiladi (5-15 daqiqa).
3. cPanel → Domains → "Force HTTPS Redirect" ni yoqing.

11-QADAM: SINOV

Brauzerda oching:

https://example.com	- bosh sahifa
https://example.com/about	- boshqa sahifa
DevTools → Network	- API chaqiruvlari to'g'ri URL'ga ketyaptimi (api.example.com)

Status code 200 bo'lsa – deploy muvaffaqiyatli.

KEYINGI DEPLOY'LAR (UPDATE)

Kod o'zgargandan keyin:

```
cd loyiha-papkasi
rm -rf .next deploy *.zip
npm run build
```

```

mkdir deploy
cp -R .next/standalone/. deploy/
cp -R .next/static deploy/.next/static
cp -R public deploy/public
cp .env deploy/.env
cp package-lock.json deploy/package-lock.json
rm -rf deploy/node_modules
zip -rq deploy.zip deploy

```

cPanel'da:

1. deploy.zip ni qayta yuklang va extract qiling (eski fayllar ustidan yoziladi).
2. Agar package.json o'zgargan bo'lsa → "Run NPM Install".
3. "Restart App" tugmasini bosing.

TEZKOR DEPLOY SCRIPT (build.sh)

Loyiha ildizida `build.sh` fayl yarating:

```

#!/usr/bin/env bash
set -e
rm -rf .next deploy deploy.zip
npm run build
mkdir deploy
cp -R .next/standalone/. deploy/
cp -R .next/static deploy/.next/static
[ -d public ] && cp -R public deploy/public
[ -f .env ] && cp .env deploy/.env
[ -f .env.production ] && cp .env.production deploy/.env
cp package-lock.json deploy/package-lock.json
rm -rf deploy/node_modules
zip -rq deploy.zip deploy
echo "OK -> deploy.zip ($(du -h deploy.zip | cut -f1))"

```

Ishga tushirish: `chmod +x build.sh && ./build.sh`

TEZ-TEZ UCHRAYDIGAN MUAMMOLAR

XATO: "application root must not contain node_modules"

YECHIM: deploy/ papkasidan node_modules'ni o'chiring va qayta zip qiling. CloudLinux uni o'zi boshqaradi.

XATO: "Cannot find module 'next'"

YECHIM: "Run NPM Install" tugmasini bosing. Yoki SSH orqali virtual env'ni activate qilib `npm install` qiling.

XATO: 502 Bad Gateway / Passenger timeout

SABAB: server.js xato berdi yoki PORT'ni eshitmadi.

YECHIM: stderr.log ni tekshiring. HOSTNAME=0.0.0.0 ekanligini tekshiring. PORT env'ni qo'lda qo'ymang.

XATO: Build "Out of memory" beradi

YECHIM: NODE_OPTIONS=--max-old-space-size=1024 npm run build
Lokaʻl kompyuterda build qiling, serverda emas.

XATO: API chaqiruvlari CORS yoki 404

YECHIM: NEXT_PUBLIC_API_URL to'g'ri ekanligini va `/api` suffiks

bor-yo'qligini tekshiring. Backend'da CORS sozlamalarini ham tekshiring.

XATO: Sahifa eski versiyani ko'rsatadi (cache)

YECHIM: Brauzerda Ctrl+Shift+R. .next/ papkasi to'liq yangilanganini tekshiring (eski fayllarni qoldirib qo'ygan bo'lishingiz mumkin).

XATO: "Image Optimization using default loader" muammosi

YECHIM: next.config.js'da images.unoptimized = true qiling yoki external loader (Cloudinary/Imgix) ulang. Standalone'da Sharp kerak bo'ladi.

XATO: ENOSPC (disk full)

YECHIM: ~/tmp, ~/.npm, eski deploy fayllarni tozalang.
du -sh ~/* bilan disk ishlatishini tekshiring.

XAVFSIZLIK ESLATMALARI

1. .env faylida MAXFIY kalit yo'qligini tekshiring (faqat NEXT_PUBLIC_ prefiksli). Maxfiy kalitlarni cPanel env panelidan qo'shing.
2. package.json ichidagi "scripts" maydonida zararli buyruq yo'qligini tekshiring (3-tomon npm paketlarini ehtiyot bo'lib ishlatilg).
3. AutoSSL ni yoqing – HTTPS shart.
4. cPanel'da "ModSecurity" yoqilgan bo'lsa, ba'zi POST so'rovlar bloklanishi mumkin. Backend alohida domenda bo'lsa, frontend'ga ta'sir qilmaydi.
5. Production'ga deploy qilishdan oldin har doim `npm audit` ni ishga tushiring va critical zaifliklarni yamang.

CHECKLIST (DEPLOY OLDDAN)

- [] next.config.js'da output: 'standalone' bor
- [] .env to'g'ri (NEXT_PUBLIC_API_URL production URL'iga)
- [] npm run build lokal'da xatosiz o'tdi
- [] deploy/ papkasida node_modules YO'Q
- [] deploy/ ichida: .next, public, server.js, package.json, package-lock.json, .env bor
- [] cPanel'da Node.js version 20+ tanlangan
- [] Application mode = Production
- [] Startup file = server.js
- [] cPanel env panelida NODE_ENV=production, HOSTNAME=0.0.0.0 qo'shilgan
- [] AutoSSL yoqilgan
- [] Backend (agar bor bo'lsa) ishlayotgani tekshirilgan

FOYDALI BUYRUQLAR (SSH)

```
# Virtual env'ni activate qilish:
source /home/USERNAME/nodeenv/myapp/20/bin/activate && cd ~/myapp

# Loglarni real vaqtda kuzatish:
tail -f ~/myapp/stderr.log

# Disk ishlatishini ko'rish:
du -sh ~/*

# Ishga tushgan processlarni ko'rish:
ps aux | grep node

# Node.js versiyasini tekshirish (virtual env ichida):
node --version

# App ni qo'lda restart qilish (cPanel ishlamasa):
touch ~/myapp/tmp/restart.txt
```

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YAKUN

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Bu qo'llanma har qanday Next.js (13/14/15/16) loyihasi uchun ishlaydi. App Router yoki Pages Router – farqi yo'q.

Asosiy qoidalar:

1. Har doim "standalone" output ishlatilg.
2. deploy paketida node_modules bo'lmasin.
3. Maxfiy env'larni .env'ga emas, cPanel panelga qo'y.
4. Birinchi marta deploy qilganda "Run NPM Install" bosing.
5. Har deploy'dan keyin "Restart" qiling.

Muvaffaqiyatli deploy bo'lsin!